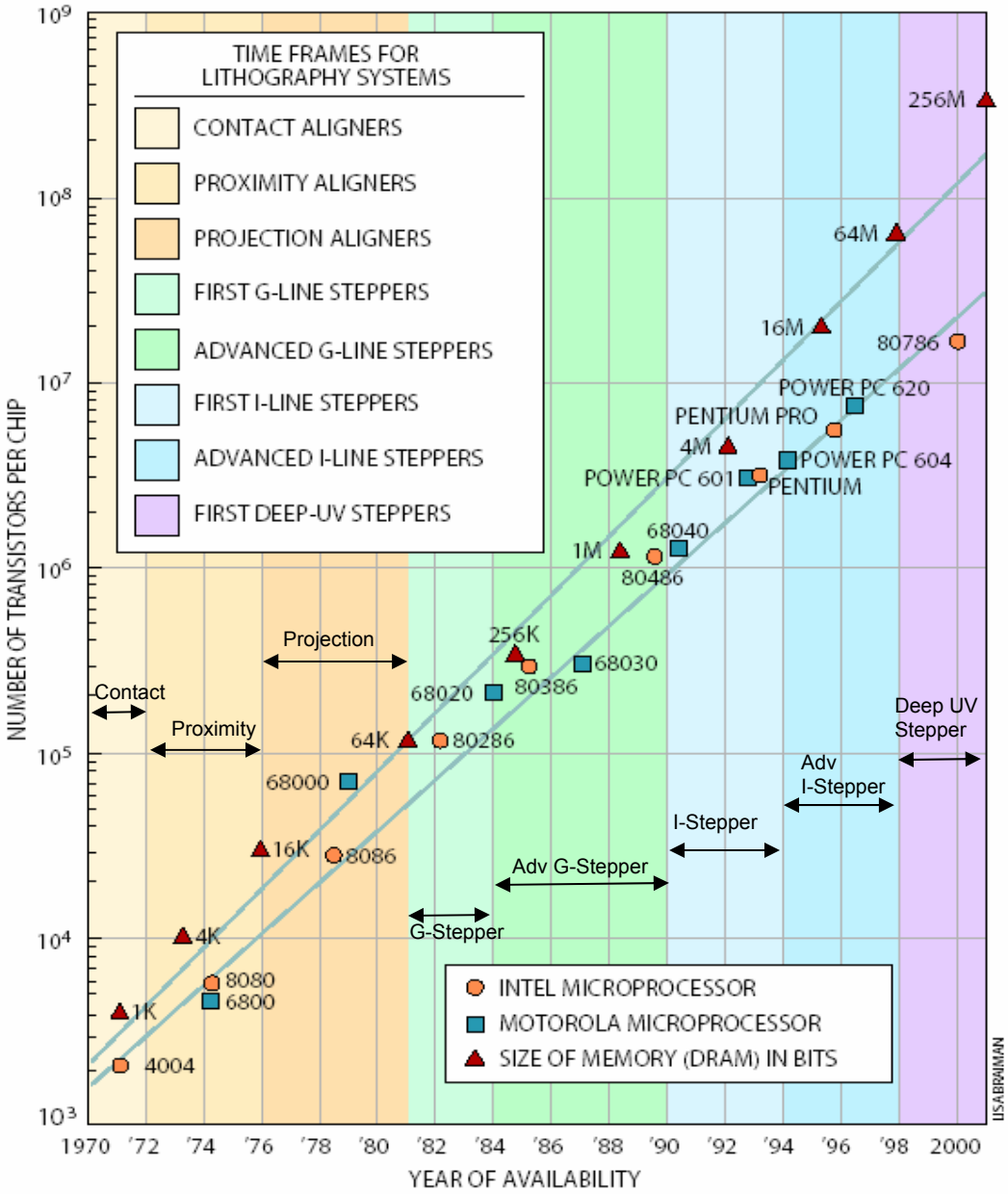
Semiconductor Technology and Moore's Law

Richard P. Rumelt



SOURCES: VLSI Research, Inc.; Integrated Circuit Engineering Corporation

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Two inventions shaped the semiconductor industry and the modern information-driven world: the transistor and the integrated circuit. Transistors were made from semiconductor materials, and integrated circuits contained thousands and millions of transistors literally etched out of semiconductor materials, all on a single "chip." By 2003, the semiconductor industry was shipping close to one billion billion (10^{18}) transistors each year. Virtually all of those transistors were contained within integrated circuits. This note describes transistors the manufacturing processes used to fabricate integrated circuits, and the mechanics of continues improvement, called Moore's Law, in the integrated circuit industry.

Transistors

In 1945, Bell Telephone Laboratories launched a research initiative to replace vacuum tube technology. William Shockley, who had joined Bell Labs in 1936 (and left during the War to develop radar for the Navy), led this effort. Shockley hired Bardeen and Brattain to aid in the project. After hundreds of trials, Bardeen, a theoretician, had a key insight: existing ideas about how electrons behaved in crystals were wrong—there were unexpected effects at the surfaces. This discovery led to the creation of the point-contact germanium transistor (Figure 1) in December 1947.[2] It was named "transistor" for "transfer resistor".

The point contact transistor demonstrated basic principles, but was not a practical device. Its performance depended upon the exact placement of contact whiskers on the germanium and was strongly influenced by surrounding conditions. Eager to show that he could do better, Shockley shut himself away in a hotel room in Chicago and, in a period of intense effort, developed the idea for the junction transistor.

Figure 1
First Point Contact Transistor

AT&T licensed the transistor for a nominal fee to other manufacturers, and waived any fees for its use in hearing aids, a recognition of the interests of the company's founder, Alexander Graham Bell. RCA, Raytheon, the newly formed Texas Instruments, and other firms began to manufacture transistors.

The Planar Process and the First IC

Shockley left Bell Labs to return to his boyhood home in Santa Clara, where he started his own company, Shockley Semiconductor Laboratories. He put together a team of renowned scientists, but the company was not commercially successful. In 1956, eight key people left Shockley Semiconductor Labs ("The Traitorous Eight"): Gordon E. Moore, C. Sheldon Roberts, Eugene Kleiner, Robert N. Noyce, Victor H. Grinich, Julius Blank, Jean A. Hoerni and Jay T. Last. Investing $3500 of their own, they started development of a double diffusion technique for manufacturing multiple transistors from a single wafer. Fairchild Camera invested $1.5 million in the team, obtaining an option to buy the company. The option was exercised in October of 1957, creating Fairchild Semi-

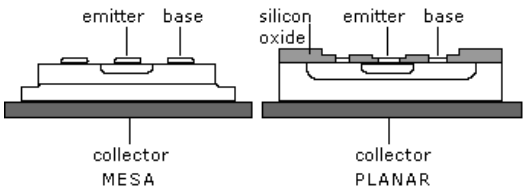
[2] Shockley, Brattain, and Bardeen received the Nobel Prize in physics in 1956 for the invention of the transistor. Bardeen received a subsequent Nobel Prize in 1972 for superconductivity.

conductor. Within six months the company was turning a profit due to an order from IBM for 100 transistors at $150 each.

Fairchild and Texas Instruments were both developing ways to produce many transistors from a single piece of germanium or silicon. These transistors would then be separated, packaged, and inserted into circuit boards where they were connected to the other critical elements of electronics circuits: resistors and capacitors. A number of people saw that it would be a great simplification if the intermediate packaging steps could be eliminated. What if transistors and resistors and capacitors could all be built on a single wafer and then connected? Such a device would be an “integrated circuit.”

Jack Kilby, an engineer at Texas Instruments, had a free summer and worked on a project of his own choosing—trying to build a whole electronic circuit out of semiconductor materials. Using mesa technology, he etched out the shapes of transistors, resistors, and capacitors in a sandwich of silicon, then connected the tops with tiny gold wires. By September 1958 he had a working device, and demonstrated it to senior management. A tiny sliver of germanium with protruding wires produced an undulating sine wave on an oscilloscope. Kilby received a Nobel Prize in 2000 for this innovation.

Figure 2



In the same year, Jean Hoerni, at Fairchild, created the planar process for fabricating transistors. Hoerni’s innovation was the methodical use of silicon dioxide as an intermediate masking material in building up the layers of doped silicon that made up a transistor. His process proved to be much more effective than the older “mesa” process with which Fairchild had begun. In the “mesa” process, layers of different types of silicon were built up on a wafer, resulting in mounds, or mesas. In the planar process, by contrast, the surface remained even and covered by a layer of silicon dioxide. Different “layers” were created by etching away portions of the silicon dioxide, treating the exposed material, then restoring the silicon dioxide surface (Figure 2).

At Fairchild, Noyce saw the connection between the planar process and the possibility of truly integrated circuits. Kilby’s approach required the intricate application of fine gold wires to connect the individual components etched on silicon. Noyce saw that the planar process could extend to the construction of metal layers, making much greater miniaturization possible. Patent disputes over the rights to the integrated circuit lasted ten years. In the end, Noyce and Kilby were recognized as joint inventors.

The power of integrated circuits was, at first, their miniaturization and lower cost per device. Use in the aerospace industry and computing industry became widespread. The first dramatic commercial success in integrated circuits was the series of TTL logic chips introduced in the mid-1960s by TI and Fairchild. Each chip contained 5-50 transistors arranged to perform specific logic functions (e.g., AND gates, 8-bit latches, etc.)

The first truly new device enabled by the IC was the small desktop calculator, developed by Kilby at TI in 1967. The ability to carry out high-precision arithmetic, the instant computation of square roots, logarithms, and powers, at one's desk was revolutionary. By the early 1970s, the hand calculator had appeared.

Figure 3
Fairchild 4100

Further developments in digital devices required advances in speed, memory, and display. Figure 3 shows Fairchild's 4100, a 256 bit RAM memory "chip" introduced in 1970. The 16x16 matrix of memory cell transistors can be clearly seen. The physical device measured about 100 mils (0.1") on each side, and was about 10 mils thick.[3] The chip was an integrated circuit containing approximately 1000 transistors on a piece of silicon only as large as the letter "o" on this page. From about this year on, most transistors would be produced as elements of integrated circuits. The single packaged transistor would appear only in special applications.

Device Innovations

Transistors were constructed out of semiconducting materials: silicon or germanium with a small amount of "doping" elements added. Adding phosphorus, for example, created p-type material and adding boron created n-type material. The basis of the transistor is the pn junction. Although p-type and n-type semiconductors are each conductors, when they are brought together the result is a device that only conducts in one direction. The conduction of electricity through the pn junction is a very subtle process and depends strongly on the chemistry of the materials and the electric fields that are present. The latter fact allowed engineers to design devices in which the flow of electricity is itself controlled by other flows of electricity or electrical charges. When the flow of electricity is controlled by other flows of electricity, one has an active device that can be used for amplification or as a switch.

The first integrated circuits were built out of *bipolar* transistors. The mechanism in a bipolar transistor was that the current (electrons) flowing between two terminals was controlled, or switched, by the current flowing into a third terminal. A small signal applied to third terminal can cause a larger, or amplified, signal to appear between the first two. Bipolar transistors were fast, but as more and more were packed into a small space there was a problem. The control current used too much power and generated unwanted heat.

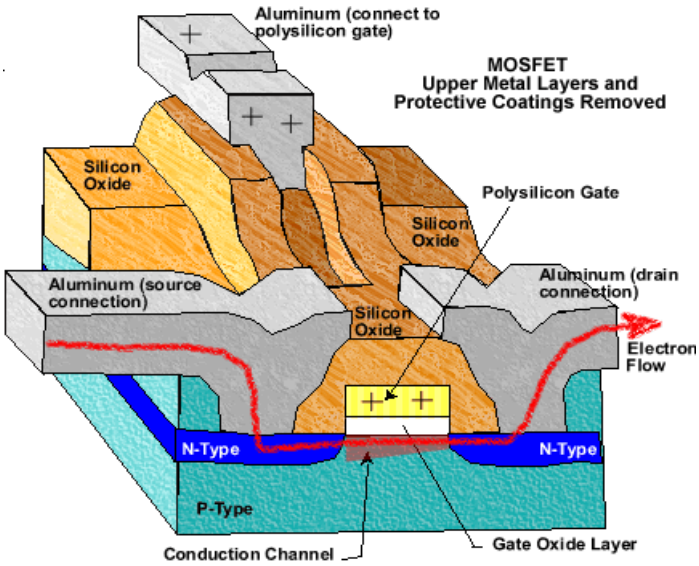
[3] A mil was 1/1000 of an inch, so 100 mils = 0.1". Just as there were 39.37 inches per meter, there were 39.37 mils per millimeter.

MOSFETs and CMOS

Another, simpler, way of creating a transistor had been understood theoretically since 1925. Instead of controlling the transistor by passing a current *into* a third terminal, an electrical charge could be applied to a "terminal" which was separated from the material by a thin insulator (Figure 4). This terminal, called the *gate*, controlled the current flowing between the other two, called the *source* and *drain*. This structure was called a MOSFET: metal oxide silicon field-effect transistor. RCA produced the first commercial MOS device in 1963.

Intuitively, the operation of a MOSFET transistor is analogous a water faucet. Water would flow from the water (source) into the sink (drain). The water can flow, however, only if the faucet is turned on. In a MOSFET, electrons can flow from the source to the drain only when the gate is charged.

Figure 4
MOSFET Transistor
(portion of an IC)



Voltage applied to the gate attracts electrons to the oxide, creating a conduction channel between the source and the drain. Turning the gate voltage off cuts the conduction channel.

MOSFET transistors could also be arranged to form logic circuits. Logic gates and logic circuits performed simple logical (Boolean) operations by coding high and low voltages as 0 and 1, or as "true" and "false." For example, the task of adding two binary numbers, say 0101 and 0110, obtaining their sum 1011, was essentially a task in "logic" and could be performed with and electronic circuit containing about 20 transistors. The circuit could be built from individual logic gates, commercially supplied as integrated circuits containing 5-10 transistors each. Or it could be created as a single integrated circuit. Or, it could simply be integrated into a much more complex device composed of thousands or millions of transistors on a single chip.

In a CMOS structure, two complementary MOS devices were created side-by-side. Although CMOS circuits tended to use more transistors, they reduced power consumption to new lows. By toggling the output between the high voltage and the low without running any current through a resistor, CMOS designs cut the power needs of integrated circuits by a factor of 1 million, making it possible to assemble tens of thousands of transistors without drawing undue power. By the 1990s, virtually all large-scale integrated circuits were CMOS.

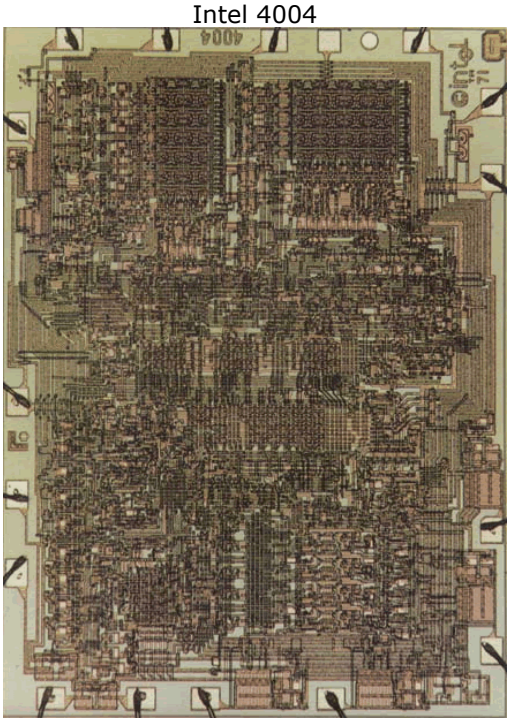
DRAM

Another key innovation was the redesign of memory cells. The traditional way to store a "bit" was the two or four-transistor *filp-flop*. Robert Dennard, an IBM engineer, realized that a capacitor, storing charge, could represent the "bit" and a single MOSFET could charge or discharge the capacitor. This concept greatly simplified the design of RAM.

Microprocessors

In 1968 Robert Noyce and Gordon Moore quit Fairchild and started a new semiconductor company—Intel. They each invested $250 thousand and raised $2.5 million in venture capital. Intel's first products were memory chips.

Figure 5
Intel 4004



When a Japanese company asked for a redesign of the chips used in its desktop calculators, Intel engineer Ted Hoff suggested that Intel build a small integrated microprocessor that would be programmable. That way, its functions could be changed by writing new software rather than redesigning the chips. Circuit designer Frederico Fagin led a team that, in 1971, delivered the Intel 4004, the world's first microprocessor.[4] It measured 1/8" by 1/16" and contained 2300 transistors. By itself, it was as powerful as the world's first digital computer, ENIAC, built in 1947, weighing tons.

A microprocessor was a "central processing unit" (CPU) built on a single chip. The microprocessor contained a few internal bytes of memory, called registers, and had connectors (pins) for input, output, and reading the contents of an external "memory" chip. It recognized some patterns of data as "instructions." By coding these instructions properly, called writing software, the microprocessor could be made to perform arbitrarily complex logical functions.

The Intel 4004 (Figure 5) was not really a new concept, as it simply integrated onto one chip the functions that had been performed by a number of individual chips. Nevertheless, it represented a break with the past, more sharply delineating the line between hardware (the chip) and software (the specific code making the chip perform a particular task). Depending upon the program, a microprocessor could run a calculator, read a keyboard, control a room temperature, control an elevator, time an engine, control a printer, and so on.

Manufacturing Overview

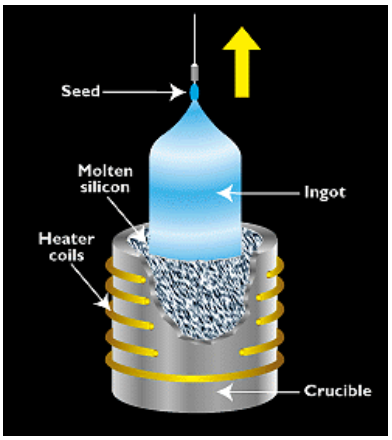
Integrated circuits were constructed by applying processes to wafers of silicon. These processes created the thousands or millions of transistors which comprised an integrated circuit. The production process was more like printing or photography than mechanical manufacturing.

The basic process was to obtain a silicon wafer from an ingot of pure silicon. The wafer was then inscribed to demark a number (usually in the hundreds) of rectangular *dice*—each individual *die* being a potential chip, itself containing thousands or millions of tran-

[4] The kit for each calculator consisted of one 4004 cpu, two 4003 serial IO chips, two 4002 320 bit dynamic RAMs, and four 4001 2048 bit ROMs.

sistors. Once fabricated and tested, the die became a chip by being mounted in a package, connected with tiny leads to external pins, and sealed.

Figure 6
Silicon Ingot



Czochralski process

Source: US. Dpt. of Energy.

Wafer sawing, lapping, polishing

Source: US. Dpt. of Energy.

Sliced Wafer

Source: Glosil Inc.

The dice were not dealt with individually. Instead, the construction of integrated circuits involved the repeated application to the entire wafer of a relatively small number of processes. The process steps created and treated layers on the wafer. The build up of these layers was controlled by the cooperative action of photolithography, which created patterns of resist on the wafer surface, and chemical processes which added material to the wafer or removed material from its surface. Early integrated circuits used about 10 layer-building, or *masking*, steps. State-of-the-art wafers might use 25-30 layers. Each layer normally took 2 to 3 days of processing time.

Silicon ingots were grown (Figure 6) from a melt of pure silicon mixed with tiny amounts of active dopants (antimony, arsenic, boron, and phosphorus). A tiny “seed” of perfect crystalline silicon was lowered into a 1400 °C melt. As the liquid silicon was gradually cooled, new crystals formed around the seed. Slowly pulling the seed into a cylinder produced the ingot of perfect crystalline silicon. The ingot was then sliced, using a diamond saw, into wafers. In 2003, most wafers were 200mm (8 inches) in diameter, with state-of-the-art wafers being 300mm (12 inches) in diameter. In 2003, an 8” wafer might be priced at $65. Adding an “epitaxial” layer of very high-quality silicon crystals might add another $25-35 to the price.

Photolithography Masks

The fabrication of integrated circuits was centered on a two-step process of masking. In the first step, photolithography was used to transfer a geometric pattern from a photomask to a light-sensitive photoresist that had been spread on the surface of the wafer. The resist was then developed, leaving a pattern of resist on the surface of the wafer that duplicated the original image on the photomask. This pattern of resist then acted itself as a mask—it protected the selected areas of the wafer from a process step such as etching, implantation, or oxidation.

Each layer required a separate mask defining the regions to be worked on that step. As feature sizes shrank, it became increasingly important to control the fine-structure of the transistors. As more transistors were added to each circuit, increasing layers of metal interconnect were required. Both forces led to an increase in the number of masking steps. By 2003, advanced designs used 22-30 masking steps.

Masks were created by computer-aided design tools. These devices used data about the integrated circuit design to control an electron-beam which drew the pattern on a resist.

Developing and plating led to a chrome mask on a glass or quartz base. In the early days of integrated circuit fabrication, the masks were applied directly to the wafer be-because any gap led to unfocused images. This *contact process* led to problems in mask wear and wafer contamination and was replaced by optical projection systems. Projection systems used a mask which contained the the wafer pattern enlarged by a factor of 4 or 5. Laser light and complex lens systems reduced and focused the image onto the wafer. The mask itself was normally mounted inside a protective transparent frame.

Figure 7
Masking Steps Applied to Part of a Wafer

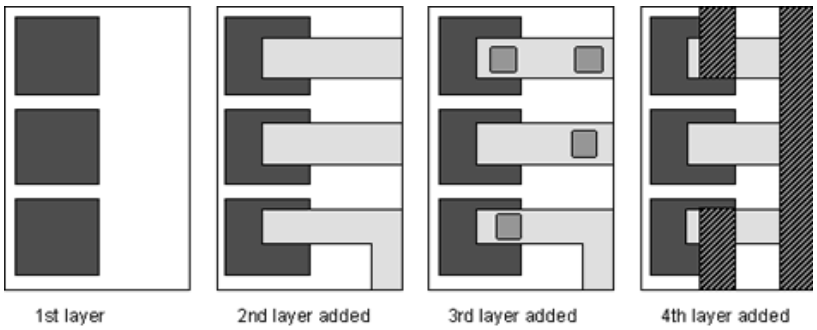
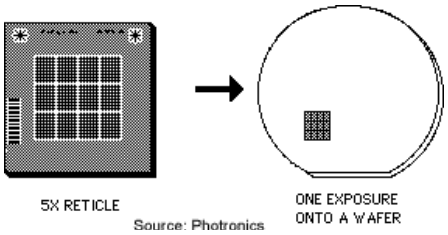


Figure 8
Reticle



Because the chips on a wafer were replicas of one another, it was not necessary for a mask to image the whole wafer. Masks imaged a number of chips in a block, called a reticle (Figure 8) and then used a *step-and-repeat process* or a *scanner* process to image the whole wafer.

The quality and accuracy of the mask was of critical importance for good results. Tiny imperfections in a mask would render all the chips produced by that image useless. Even if each mask was perfect, their overlaid images had to match for the build-up of 20 or more masking steps. Finally, the lithography equipment had to align each image accurately, with tolerances of a fraction of a wavelength of light.

The cost of a set of reticles depended upon the feature size being imaged. In 2003, a typically state-of-the art mask set for 130nm technology might require 16 “critical tolerance” reticles, costing $32-38,000 each, and 10 “non-critical” reticles, costing $22-28,000 each. Each reticle had a life of about 4000 exposures (less than one week) before it had to be replaced. The cost per wafer processed of this whole set was, therefore, on the order of $100.

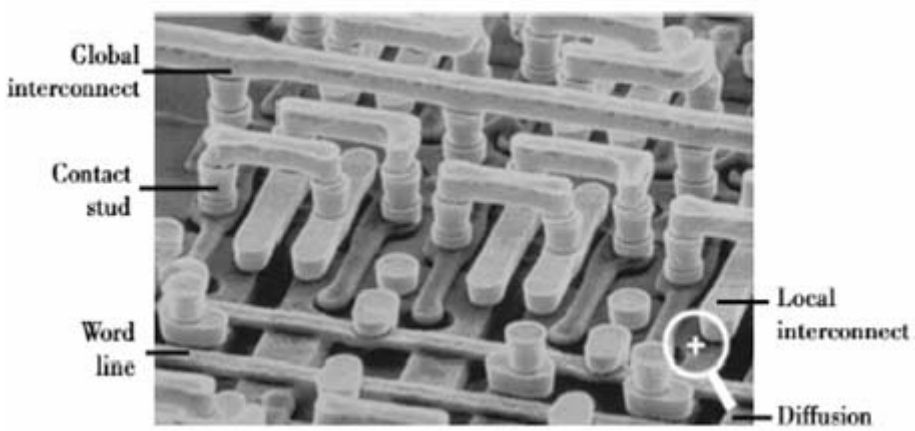
Fabrication Process

The fabrication process is diagrammed in Exhibit 1, which shows the steps required to construct a pair of (CMOS) transistors. As can be seen, the process involved the repeated application of photolithography to isolate certain regions, then the application of treatments to those regions. State-of-the-art chips were built by combining hundreds of steps. The exact steps carried out and their precise sequence were often closely held trade secrets.

The metal interconnects among transistors on a chip were also constructed with these processes. As engineers integrated more transistors onto logic chips, more complex

problems of interconnection arose. By 2003, complex logic chips contained 5 to 7 layers of metallic interconnects. Figure 13 shows a microphotograph of the interconnects on a DRAM chip where all the protective oxide has been etched away to reveal the metal conductors. Each of the four layers of metal required several process steps, as did the plugs which acted as vertical connectors among levels.

Figure 13



Source: IBM

Yield

After a wafer was processed, it was probe-tested. The purpose of this step was to ascertain the quality of each die on the wafer before undertaking the costs of more handling, shipping, and packaging. The probe-tester was a machine which held the wafer and stepped an electronic test probe over the dice. At each die, tiny needles were pressed onto the edge-connectors of the die and the integrated circuit was electronically tested. The percent of chips passing this test was called the *wafer-probe yield*. This yield was a key determinant in the cost structure of integrated circuit manufacturing.

Costs for DRAMs and microprocessors are estimated in Exhibit 2. A 1 Mbit DRAM chip contained about 1 million transistors. In 1988, the cost of processing a 150mm wafer (including the allocation of fixed overhead) was about $400. Each wafer contained, in this example, 546 dice (potential DRAM chips), giving a total processed cost per die of 400/546 = $0.73. However, the wafer-probe step, which tested the dice, had a yield of only 67%, so that the cost per good die rose to 400/366 = $1.09. Placing the chip into a package and connecting it to the external pins or balls (packaging) plus the final test steps added $0.31 in cost per die processed and resulted in the discard of another 44 dice per wafer. The final cost per chip was $1.59.

Figure 14

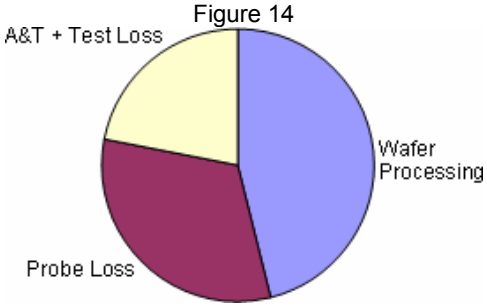


Exhibit 2 shows the relative importance of wafer processing, loss to probe-testing, and assembly, final test, and final test loss. Thirty-two percent of the final product cost was due to yield losses at the wafer-probe step.

On start-up, yield rates in the probe-step could be as low as 5-10%. Most fabs hoped to get yields up into the 90-98% range by the time their processes had "burned in." Nevertheless, the industry showed a great deal of variance in yield rates. The Berkeley benchmarking study[5] showed DRAM yields varying from 35 to 90% across multiple fabs and production lines in 1997-98. Most of the data clustered within the 85-95% range.

Defects in dies had, historically, had three major sources: (a) 2% from defects in materials, including the wafer itself, (b) 8% from processing errors, including photolithography and mask defects, and (c) 90% from contamination.

Contamination meant unwanted particles adhering to the wafer or mask, causing defects to be etched or deposited onto the patterns on the wafer. If a defect was large enough, and fell in the wrong place, it would cause a critical failure. At the simplest level, two conductors might be improperly connected, shorting the circuit, or a connector might be interrupted, breaking a circuit. More subtly, device performance might be adversely affected by a defect, so that its operation was slow, temperature sensitive, or erratic.

In turn, contamination came from air particles, human presence, equipment, and processes. During the early years of the integrated circuits industry, a great deal of effort was expended on cleanroom technology—the methods for reducing particulate matter. Cleanroom practice started with HEPA air filters and added positive pressure control, so that uncleaned air did not flow into the area. To this were added special clothing and handling systems. The old cleanroom standard classified cleanrooms according to the number of particles 500nm and larger per cubic foot. Thus, a class 1 cleanroom had only 1 particle per ft^3, whereas a class 100 cleanroom had 100 particles.

According to the new U.S. Federal Standards, if the ISO class of a clean room was the integer N, then there had to be fewer than 10^N airborne particles per cubic meter having a size of 100nm or larger (Table 3). Both systems of measurement were in use.

Studies had shown that a person sitting still, properly garbed, emitted 100,000 particles per minute of size 0.3 mm or larger. Simple movements, like toe tapping, raised the rate of emission to 1,000,000 particles per minute! Thus, in a class 1 cleanroom, the limiting factor became the number of people per square-meter and the time the wafer was exposed to them.

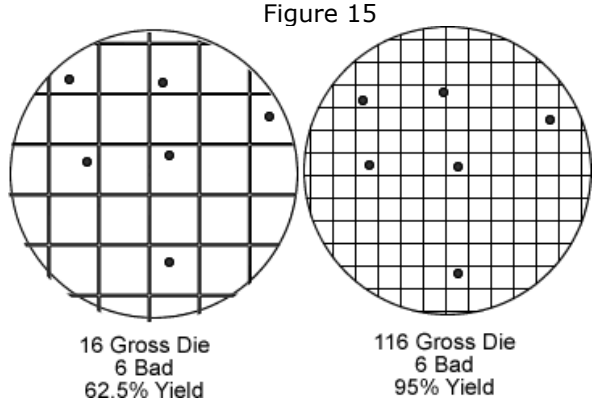
The cost of cleanrooms rose dramatically with the stringency of its ratings. Class 100 cleanrooms cost about \$1200 per ft^2 in 2002, whereas a class 1 cleanroom cost upwards of \$10,000 per ft^2. A class 0.5 cleanrooms cost \$25,000 per ft^2 and more.

[5] Robert C. Leachman, "Competitive Semiconductor Manufacturing: Final Report from Benchmarking Eight-inch, sub-350nm Wafer Fabrication Lines," UC Berkeley, 2002.

Each wafer was divided into a number of dice. Each die was a potential chip (e.*g.,* a DRAM or a microprocessor, a digital signal processor, a graphics controller, or a custom logic device.)

The gross yield was the number of dice on the wafer testing "good" divided by the total dice on the wafer. As can be seen in Figure 15, the larger a die, the more chance it had of encountering a particles of a killer size or larger. Much effort had been expended to quantify this association.

Figure 15



The simplest yield formula used in the industry was based on a model in which defects were distributed randomly on the wafer. Letting the expected number of killer defects per mm^2 be D (the defect density) and letting the active area of a die be A_d mm^2, then the expected number γ of killer defects on one particular die was

$$\gamma = A_d D \,. \tag{1}$$

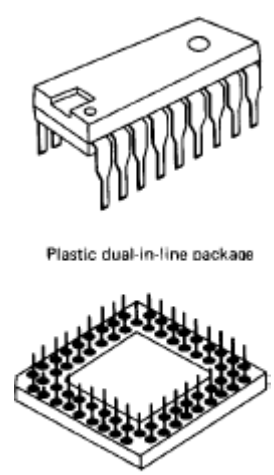
The probability of failure is the probability that $\gamma \geq 1$. The standard assumption was that (1) represented the mean of a gamma distribution. Integration gave this formula for yield (the probability that $\gamma = 0$):

$$Y = \left(1 + \frac{A_d D}{\alpha}\right)^{-\alpha} = \left(1 + \frac{\gamma}{\alpha}\right)^{-\alpha} . \tag{2}$$

Here, α was the "clustering" parameter and reflected the degree to which particle positions were correlated. Normally, α had a value in the neighborhood of 2, but ranged from 0.3 to 5.

Equation (2) implied that yield falls as the area of a die increases. Conversely, yield rises as the area of a die decreases (other factors being constant).

Figure 16
Traditional Packaging



Test and Packaging

Once the wafer had been probe-tested, the individual dice were sawed apart using a diamond blade. Each die was mounted in a frame which became an integral part of the final package. At the edge of each die were bond pads which were its points of connection. Connections were made between these bond pads and spots on the frame that connected to the external pins.

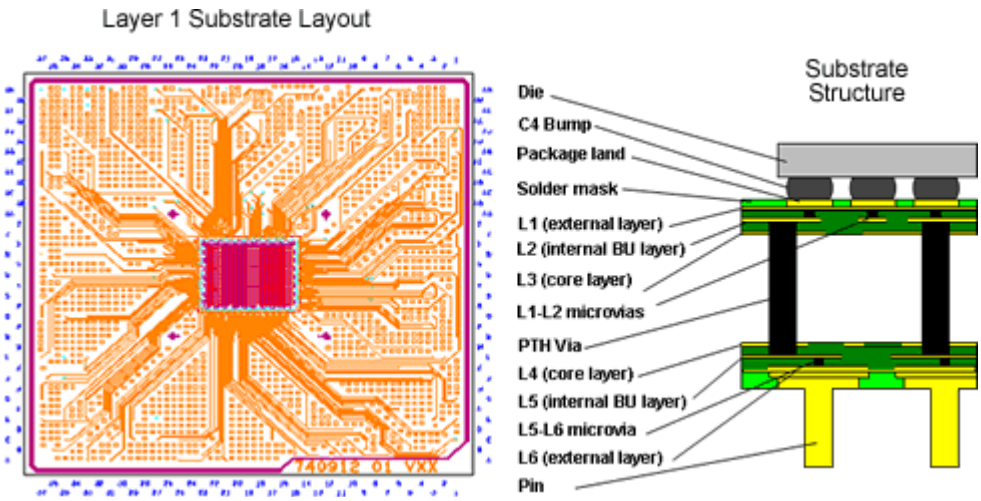
As chips became more complex, more pins were required to attach to external circuits. In part, more power pins were required so that each part of the device was fed power, preventing internal lags. In addition, the integration of more functions and wider data buses (16 bit to 32 bit to 64 bit) meant more pins. In general, the number of

pins on a logic chip tended to increase as the log of the number of transistors. However, sufficient modularity could break this rule of thumb.

In the mid 1990s, new styles of packaging arose to meet the needs of denser pin structures. Instead of having pins just on the edges of a chip, the whole undersurface of the package was covered with pins (pin grid array).

As densities increased, the creation of a connection between the integrated circuit and the outside world became more complex. To act as an intermediary between the scale of the chip and that of the circuit board, substrates became more complex. The substrate was like a miniature circuit board, handling the wiring from the chip bond pads to the pins or solder balls that are its external connections. By 2003, substrates were constructed with lithographic techniques to fine tolerances. The connections between the substrate and the outside circuits was made with either pins or a ball grid array (BGA).

Figure 17
Intel CopperMine Substrate



Source: Intel

BGA packages were directly connected to a dense circuit board or to another large substrate which integrated two to five BGA chips. These new structures provided a level of integration intermediate between the IC and the circuit board. Many observers noted that as feature sizes became smaller, more effort and cost went into packaging and connections.

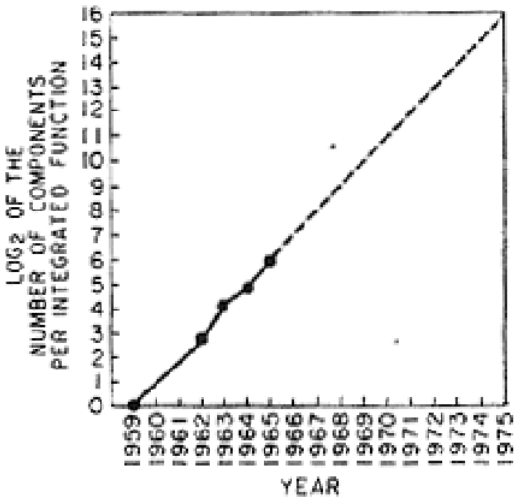
Moore's Law and Scaling

The semiconductor industry was unique in the progress it had achieved since its inception. The cost reduction and performance increases had been continuous and phenomenal. Only the data storage industry had scaled so dramatically in so short a time.

Gordon Moore was first asked to comment on the probable evolution of integrated circuits in 1965,[6] when he was Fairchild's Director of R&D. He wrote:

> Reduced cost is one of the big attractions of integrated electronics, and the cost advantage continues to increase as the technology evolves toward the production of larger and larger circuit functions on a single semiconductor substrate. For simple circuits, the cost per component is nearly inversely proportional to the number of components, the result of the equivalent piece of semiconductor in the equivalent package containing more components. But as components are added, decreased yields more than compensate for the increased complexity, tending to raise the cost per component. Thus there is a minimum cost at any given time in the evolution of the technology. At present, it is reached when 50 components are used per circuit.…The complexity for minimum component costs has increased at a rate of roughly a factor of two per year…. That means by 1975, the number of components per integrated circuit for minimum cost will be 65,000. I believe that such a large circuit can be built on a single wafer.

Figure 2
Moore's 1965 Forecast



Note that Moore did not say that progress was measured by the density of transistors per unit area, or by the cost of a transistor. Here, he argued that *the number of components that could be integrated onto one chip depended upon a balance between the savings from "cramming" and the costs of decreased yield*. The size and component count of a chip, therefore, was an engineering and business decision that reflected the state of the technology. Progress could be measured by the number of components that could be squeezed onto a chip. Based on three or four data points, he forecast an annual doubling (Figure 2).

Moore's 1965 paper has been based on experience and a general faith in the industry's ability to make devices ever smaller. In 1972, Dennard and Gaensslen of IBM developed a theory of MOSFET scaling. Their surprising result was that if the electric field strength over the gate were was held constant, simple reduction in scale of a MOSFET would work, and would produce improved performance on almost every dimension.[7] In particular, if the linear dimension of the transistor were cut in half, then the voltage necessary was cut in half and the current flowing was cut in half. The power consumed was cut by a factor of four. The transistor density (number per unit area) was increased by a factor of four. Furthermore, the frequency of operation was doubled. (In addition, the doping concentration needed to make the device work doubled.)

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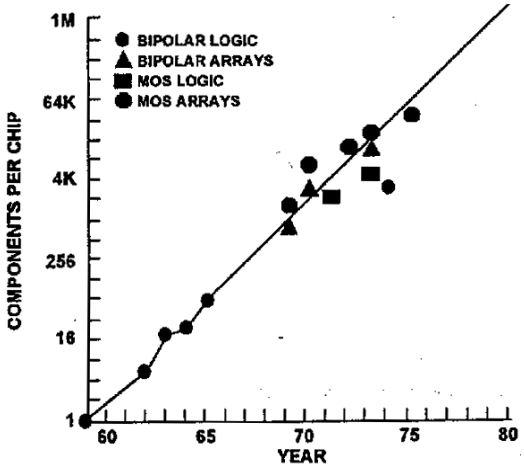
[6] G.E. Moore, "Cramming More Components onto Integrated Circuits, Electronics, 1965.

[7] Reported in Dennard, et al., "Design of Ion-Implanted MOSFET's with Very Small Physical Dimensions," IEDM, 1974.

Figure 3A
Moore's 1975 Data



With the development of the scaling rules, CMOS technology had huge advantages: it used much less power and could be simply scaled. With these understandings in hand, the progress of integrated circuits was paced by process technologies rather than device design.

In 1975, by then a founder of Intel, Gordon Moore gave a presentation in which he reviewed his earlier forecast. He had predicted chips containing 65,000 transistors in 1975. A new charge coupled memory device about to be released by Intel stored 16 Kbits. Moore's drawing (Figure 3A) placed this device at the 32,000 component level of complexity. In his analysis he treated it as if it had 64,000 components.[8]

Moore's analysis made these points:

- The component count for chips introduced since 1965 "fit pretty well" the original forecast of "doubling every year." (Figure 3A)
- Part of the increased transistor count was simply due to larger die (or chip) sizes. At constant density, the number of transistors on a chip was proportional to the die's area. Chip sizes had increased by about a factor of 20 in 16 years.
- Part of the increased transistor count was due to higher density—more transistors per unit area. The transistor density was inversely proportional to the square of feature size, so that as feature size fell, density increased.[9] Examining feature size trends, Moore argued that feature size reductions had created a 32X increase in transistor density between 1961 and 1975.
- Putting the above three factors together, Moore could only account for a portion of the increase in transistor count he saw between 1959 and 1975 (Figure 3B). He attributed the remainder to "circuit and device cleverness." In particular, he credited the advent of MOS devices and better isolation to greater packing densities.

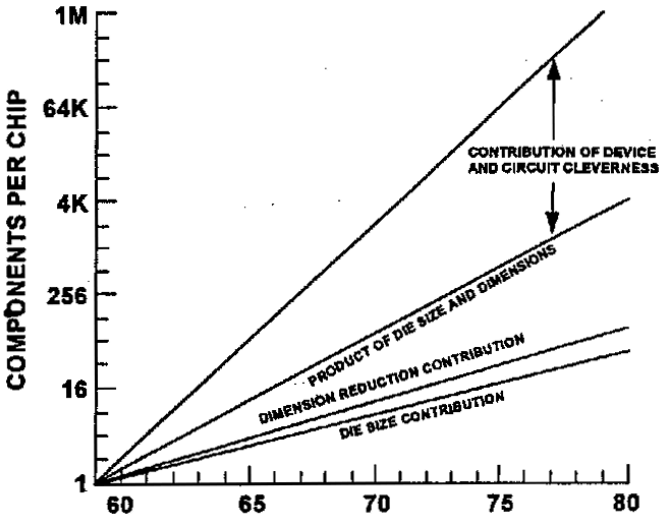
Looking ahead, Moore forecast that there would be less and less opportunity for "cleverness." The complexity (component count) curve would slow down, he forecast. Within five years, the rate of progress would fall to a "doubling every two years."

[8] In his 1965-75 papers, Moore spoke of "components," which may have included transistors, capacitors, resistors, etc. By 1979 and thereafter, the industry spoke of transistors rather than components. A CCD memory was a then promising technology which was eclipsed by DRAM. In a CCD memory, bits were constantly shifted along a chain of overlapping "transistors." Since the structure combined transistors, overlapping gates, and capacitors in a whole, it is difficult to assign a component count. Intel's 16Kb memory chip would have had 16,000 overlapping transistors and one could count another 16,000 capacitors, but that is not the practice in assessing DRAM.

[9] Feature size was normally limited by cleanliness and lithography, and was usually measured by the minimum distance between parallel "lines" on the IC.

Once again, in 1979, Moore revisited the issue of the progress in integrated circuit complexity.[10] He restated his view that complexity of the most advanced chips had been doubling each year during the first 15 years of the industry and subsequently doubling every two years. However, he added some new observations. In particular, Moore

Figure 3B



noted that design costs were coming to dominate manufacturing costs. Design costs, he estimated, were inflating at 10 percent per year. In addition, there were increasing problems in on-chip connections and packaging that limited the gains to improved transistor density. The rising costs of designing increasingly complex chips would have put a quick stop to progress in integrated circuits, he argued, if the calculator and memory not appeared:

> In general, the semiconductor industry's efforts to solve its problems in the 1965-1968 era were not successful. The product definition crisis persisted and limited IC complexity through the mid-60s. Two things broke the crisis for semiconductor component manufacturer, though not necessarily for the mainframe computer manufacturer: the development of the calculator and the advent of semiconductor memory devices.

Thus, Moore credited the microprocessor and associated memory devices with the continuing progress of the industry. Without these products, demand would have been limited to the sale of individual logic components or to the very expensive integrated chips needed by mainframes and aerospace applications. In essence, the microprocessor offloaded the cost of logic design onto software engineers, and allowed semiconductor manufacturers to reap economies of producing standardized chips. Microprocessors, in turn, needed memory. The exploding demand for microprocessors and memory were the dynamo that drove “Silicon Valley.”

Moore’s projection of exponential growth in transistor count per chip was dubbed “Moore’s Law” by Carver Mead, a Cal Tech semiconductor physicist. Moore’s Law was

[10] G.E. Moore, “VLSI: Some Fundamental Challenges,” *IEEE Spectrum*, 1979, pp. 30-37.

very widely, and incorrectly, stated as the rule that chip density will double each 18 months. As Moore himself noted,

> I never said 18 months. I said one year, and then two years. One of my Intel colleagues changed it from the complexity of the chips to the performance of computers and decided that not only did you get a benefit from the doubling every two years but we were able to increase the clock frequency, too, so computer performance was actually doubling…. Moore's Law has been the name given to everything that changes exponentially in the industry…. if Gore invented the Internet, I invented the exponential.[11]

Whatever Gordon Moore originally meant, or said, the "double every 18 months" rule became a touchstone in Silicon Valley. As one industry watcher put it[12]:

> Moore's Law is important because it is the only stable ruler we have today. It's a sort of technological barometer. It very clearly tells you that if you take the information processing power you have today and multiply by two, that will be what your competition will be doing 18 months from now. And that is where you too will have to be.

The semiconductor industry's chief coordinating instrument was the International Technology Roadmap for Semiconductors (IRTS), produced by an international working group. This complex document provided a coordinated forecast of the detailed technological characteristics of integrated circuits up to 15 years into the future. Specifically, the IRTS defined a new technology "node" as occurring when there was a 0.7X reduction in feature size (0.5X increase in density). This reduction in feature size was called *scaling*. IRTS claimed that during 1974-1999 the node cycle time was 2 years. It projected future nodes to occur every three years. Since its start in 1992, the IRTS made the assumption that

> continued scaling of microelectronics would further reduce the cost per function (averaging ~25% per year) and promote market growth for integrated circuits (averaging ~17% per year). Thus, the Roadmap has been put together in the spirit of a challenge—essentially, "*What technical capabilities need to be developed for the industry to continue to stay on Moore's Law and other trends*?"

The IRTS Roadmap was studied by everyone in the industry. For the challenges to be met, simultaneous investments in solving problems had to be made by mask-makers, process equipment designers, wafer producers, resist manufacturers, lithography equipment producers, device designers, test and packaging engineers, etc. The Roadmap provided not only a forecast of product characteristics, but a fairly detailed explanation of the technical challenges in each supplier area that would have to be met to keep the industry on the trend-line.

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[11] D.J. Yang, "On Moore's Law and Fishing: Gordon Moore Speaks Out," *U.S. News Online*, 2000.
[12] Malone, Michael S. 1996. "Chips Triumphant," *Forbes ASAP*, February 26, pp. 53-82.
.

Despite the mechanical aspect of the Roadmap forecast, many experts believed that Moore's Law was more of a social than a physical phenomenon. Carver Mead offered this explanation of how Moore's Law actually worked:

> After it's [Moore's Law] happened long enough, people begin to talk about it in retrospect, and in retrospect it's really a curve that goes through some points and so it looks like a physical law and people talk about it that way. But actually if you're living it, which I am, then it doesn't feel like a physical law. It's really a thing about human activity, it's about vision, it's about what you're allowed to believe. Because people are really limited by their beliefs, they limit themselves by what they allow themselves to believe what is possible. So here's an example where Gordon, when he made this observation early on, he really gave us permission to believe that it would keep going. And so some of us went off and did some calculations about it and said, 'Yes, it can keep going'. And that then gave other people permission to believe it could keep going. And [after believing it] for the last two or three generations, 'maybe I can believe it for a couple more, even though I can't see how to get there'…. The wonderful thing about it is that it is not a static law, it forces everyone to live in a dynamic, evolving world.[13]

The Mechanics of Moore's Law

The increasing complexity of integrated circuits was driven by a constant reduction in the size of transistors and other components. Because of the scaling rules, designers kept track of the smallest feature that had to be carefully etched—calling it *the feature size*. Between 1971 and 2003, the feature size of transistors fell from 10 microns[14] to 0.1 microns—a factor of 100 (Figure 5).

Other things being equal, a reduction in feature size by a factor of 100 meant that transistor density (number of transistors in a given area) increased by a factor of 100X100 = 10,000. That is, if other things were equal, a chip that held 3000 transistors in 1971 could hold 30 million in 2003.

The relative constancy in the cost per square cm for wafer processing had been achieved in spite of the increasingly complex processes employed. This increased process complexity had been largely offset by increases in the diameter of the wafers used. Whereas the original Intel 4004 had been produced on a 2" (50 mm) wafer, 6" wafers were commonplace in the mid-1980s. By 1998, the bulk of industry production was on 8" (200 mm) wafers, and many firms were planning the conversion to 12" (300 mm) wafers.

If the MOS scaling rules had applied perfectly, and had the cost per cm^2 of silicon remained constant, down-scaling transistors by a factor of 100 would have reduced the cost per transistor by a factor of 10,000, reduced power consumed per transistor by a

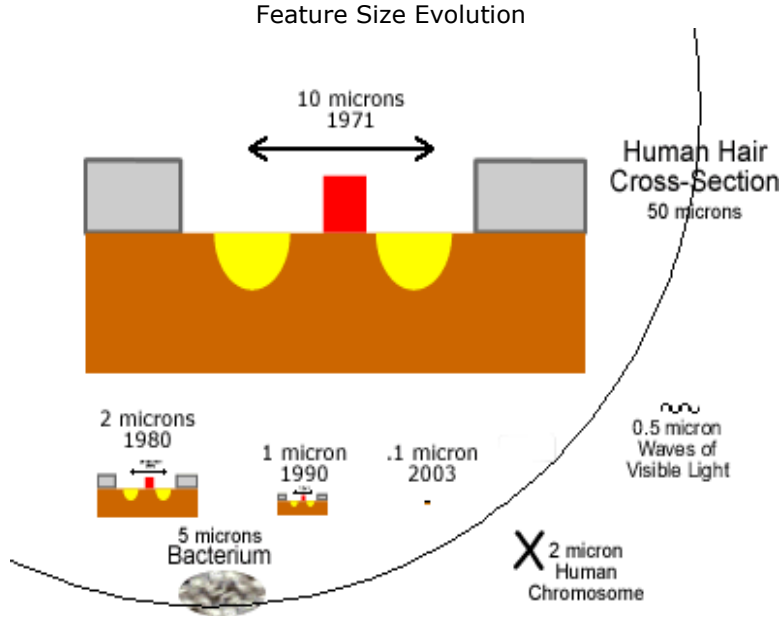
[13] Carver A. Mead 1992.

[14] A micron is one-millionth of one meter (one ten-thousandth of a millimeter) abbreviated as μm or simply μ. There are 1000 nanometers in one micron, so 0.1 μm = 100 nm.

factor of 10,000, and increased operating frequency by a factor of 100. Although technology buffs liked to point to each of these general trends as if they were separate technological achievements, they were all a direct result of scaling.

The scaling of devices, the continued reduction in feature size, pushed the semiconductor industry towards amazing improvements in complexity and price. Exhibits 5 through 8 display data on the historical trends in microprocessor price and performance, and DRAM prices.

Figure 5
Feature Size Evolution



Pushing Lithography—Resolution Enhancement Technology

Optics is the science of light. The principles of lenses, imaging, and how light waves interfere with one another were well understood since the late 19th century. A central principle of optics was the criterion, developed by Lord Rayleigh, for the resolving power of an optical system. Rayleigh's criterion was developed by considering the *diffraction* patterns of light. When light from a point source is focused by a lens, the image is not a perfect point. Instead, of being a point, or a simple disc, the image produced is a set of concentric rings of light and dark called an Airy Disc. In simple terms, small features of an image are blurred because the lens is finite.

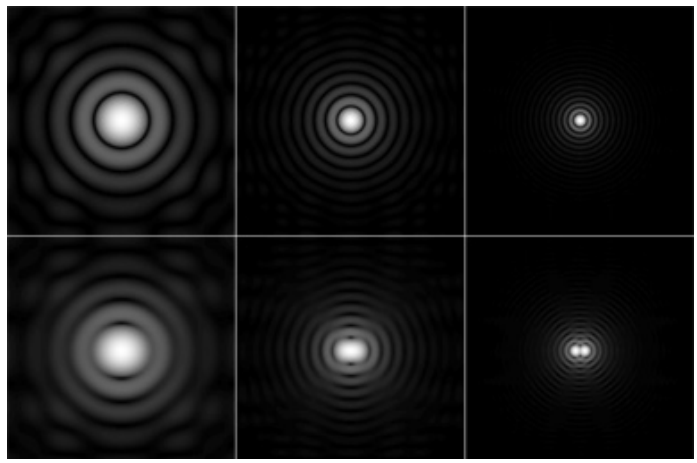
Figure 6 shows the Airy Discs produced on photographic film by one point source (above) and two points (below). As the size of the lens is increased, the Airy Disc becomes more compact. With a sufficiently large lens (right) the Airy discs become small enough to resolve the two points. Rayleigh's Criterion explained why a larger lens gave a sharper image. Larger telescopes and binoculars showed more details. Rayleigh's Criterion also explained the resolving power of the human eye and helped define 20/20 vision.

Replacing the problem of two stars with that of resolving two lines on a mask into two lines on a wafer, Rayleigh's Criterion becomes:

$$F = \frac{k_1 \lambda}{n \sin \alpha} = \frac{k_1 \lambda}{NA}$$

where F is the feature size, λ is the wavelength of light forming the image, n is the index of refraction of the medium (n = 1 for air), α is the capture angle of the lens (light gathering ability) and k_1 is a measure of the "complexity" of the photolithography system.

Figure 6
Airy Disc's of Single (upper) and Binary (lower) Stars for Increasing Lens Diameters



An optical system's *numerical aperture* was defined as NA = $n \sin \alpha$ and was never larger than n. Similarly, physical principles required that $k_1 \geq 0.5$. Together, these constraints implied $F \geq \lambda/2$. Thus, laws of optics seemed to prohibit lithography of features separated by anything less than ½ of a wavelength of light. More practically, real-world optical systems tended to have NA = 0.75 and k1 ≈ 0.6, so the realistic spacing limit was closer to 1 wavelength of light.[15]

Normal light used in lithography had a wavelength of 0.5-0.6 μm. The idea that physics limited photolithography to feature sizes of ½ micron and larger was generally accepted in the early 1980s. In 1983 two Bell Laboratories scientists stated that

> After consideration of all factors which limit resolution such as exposure hardware, resist systems, registration, alignment, and line width control, there is a general consensus that the useful resolution limit of photolithography may lie somewhere between 0.4 and 0.8 um and depends on such factors as the implementation of short wavelength UV and the ability to accurately place images from a projection tool onto a silicon wafer.[16]

Some experts were echoing this view into the early 1990s. For example, Tennant and Hielmieier[17] wrote: "Most experts agree that the exponential decrease in the minimum line width of optical lithography (the current method for "printing" circuits on silicon) will probably reach a limit of about .25 micron in the 1990's...minimum line widths for high-volume ICs will not get much smaller than .25 micron."

[15] The limitation is often stated as restricting the minimum feature size. This is not strictly true. By using a photoresist that is responsive to only the most intense light, a single feature of arbitrarily small size can be imaged. However, the Rayleigh Criterion limits the minimum *spacing* between two features.
[16] L. F. Thompson and M. J. Bowden, "The Lithographic Process: The Physics," in Introduction to Microlithography : Theory, Materials, and Processing, Thompson and Bowden, eds., based on a workshop sponsored by the ACS Division of Organic Coatings and Plastics Chemistry at the 185th Meeting of the American Chemical Society, Seattle, Washington, March 20-25, 1983.
[17] Harry Tennant and George H. Heilmeier, "Knowledge and Equality: Harnessing the Tides of Information Abundance," in Technology 2001: The Future of Computing and Communications, Derek Leebaert (Editor), Cambridge: The MIT Press, 1991.

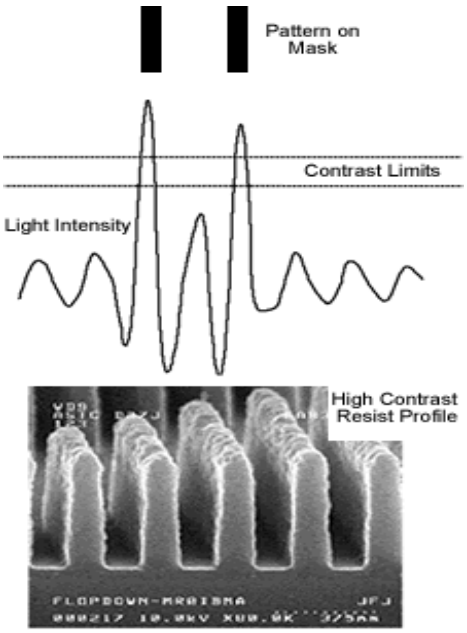
Given these hard limits, it seemed clear to many that the industry would have to push beyond visible light and use shorter wavelength ultraviolet light or even x-rays, eventually moving to electron-beam lithography. The problems with such moves were immense—most materials were opaque to ultraviolet light, including class and quartz. Lenses would have to be replaced by focusing mirrors. Additionally, whole new classes of resists would have to be developed.

Despite considerable investment in x-ray and electron-beam techniques, there were a number of engineering developments that confounded the predictions of physicists. Three developments—*high-contrast* photoresists, *optical proximity correction* (OPC) and *phase-shift masks* (PSM)—were the most important. All broke the hard logic of the Rayleigh Criterion by changing the problem.

High-Contrast Photoresists

Consider the problem of imaging a series of rectangular lines. If the distance between the lines was on the order of a wave length of light, diffraction would make the image of the lines blurred and indistinct and the spaces between them would not be reliably reproduced. However, a very *high contrast* photoresist could still produce sharp-edged lines after development! Figure 7 displays this phenomenon, showing how very sharp edges in high-contrast resist could be recovered even when imaging rectangles on a 250 nm pitch with 248 nm light.

Figure 7
High Contrast Photoresist Adds Sharpness



The key to this effect was the *chemically amplified* resist, developed through collaborative work between IBM and universities.[18] In this system, light creates a *catalyst* which drives a second chemical reaction during the bake phase. The amplification occurs in that one photon of light leads to alterations in many atoms rather than just one. This ratio is called the quantum efficiency, and chemical amplification pushed it above unity. Commenting on the power of the new photoresists, Lloyd Harriott, head of advanced lithography at Lucent Technology's Bell Labs, said "The chemists saved the day. Improvements in deep UV photoresists were so high-contrast you could pull out images better than people thought."

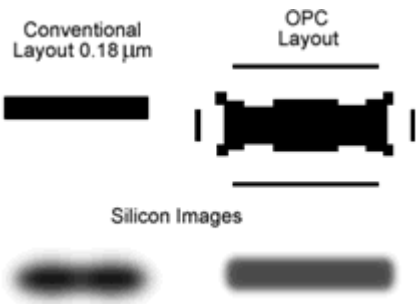
Ultimately, this technique worked because the patterns being imaged were binary: the desired image was either "on" or "off" at each position, there was nothing "soft" in the pattern to be imaged. The negative aspect of these new chemicals was their extreme sensitivity to low concentration chemicals in the air due to paint, dust, etc.

[18] Especially Grant Willson at IBM, now at the University of Texas, Austin and Jean Frechet, University of Ottawa and UC Berkeley.

OPC

Optical proximity correction (OPC) methods were first developed in the 1970s, but were not commonly used until the 1990s. The traditional way of projecting a rectangular shape onto the resist was to build a mask with a rectangular pattern on it and project its image. However, as the size of the rectangle became comparable to a wavelength of light, the resultant image was blurred (Figure 8). However, by understanding and predicting the effects of diffraction, engineers could build a mask with a non-rectangular pattern which would "blur" into a rectangle! The left panel of Figure 8 shows what happened when a 180 nm feature was submitted to 250 nm technology; the result was a blurred image with the center section necked-down. By altering the mask, however, a perfectly presentable 180 nm feature could be produced with 250 nm technology.

Figure 8



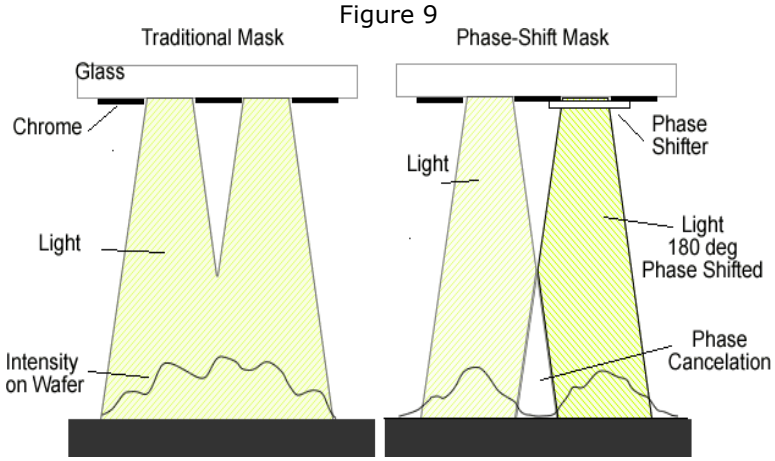
PSM

As engineers began to work with this idea, they also began to adopt *phase-shift mask* (PSM) techniques. Theoretically understood since 1982, these methods were developed by leading Japanese mask suppliers, working in cooperation with their DRAM customers. This technique could produce even finer features than OPC, but was much more complex to implement.

Phase-shift masks were built out of more than one kind of material, say glass and quartz. Instead of controlling diffraction patterns with specific shapes, as in OPC, phase-shift methods used the differing thicknesses of materials to induce sharper corners and edges in the image.

Figure 9 shows the operation of a PSM. Light representing one feature shines through two slits in the mask. The distance between the slits is less than 1 micron, producing wafer images less than 0.25 micron apart. When the right slit has an additional quartz thickness added, the light emerging is 180 degrees out of phase with the light from the left slit. Light that is perfectly out of phase cancels, creating a dark area where the light from both slits overlap. Thus, the PSM creates two separate images rather than one blurry composite.

Figure 9



RET methods reduced the effective or implied value of k_1 being achieved below the "theoretical" limit of ½. Classical optics had predicted that 248 nm light could not create features of less than 124 nm, even using perfect equipment. Nevertheless, by 2003, 248 nm light waves were creating image features smaller than 120 nm on a production basis. Still, most observers believed that it would not be possible to push k_1 below ¼. If that were true, and if Moore's law was to continue, optical lithography would be left behind in 2007. Some firms were again investing considerable funds in next-generation lithography. Intel led a consortium pressing for 13 nm soft x-ray (extreme ultra-violet) technology. IBM was advancing electron-beam methods. Other technologists, however, believed that there really were no limits to what could be done with optical lithography and cleverness.

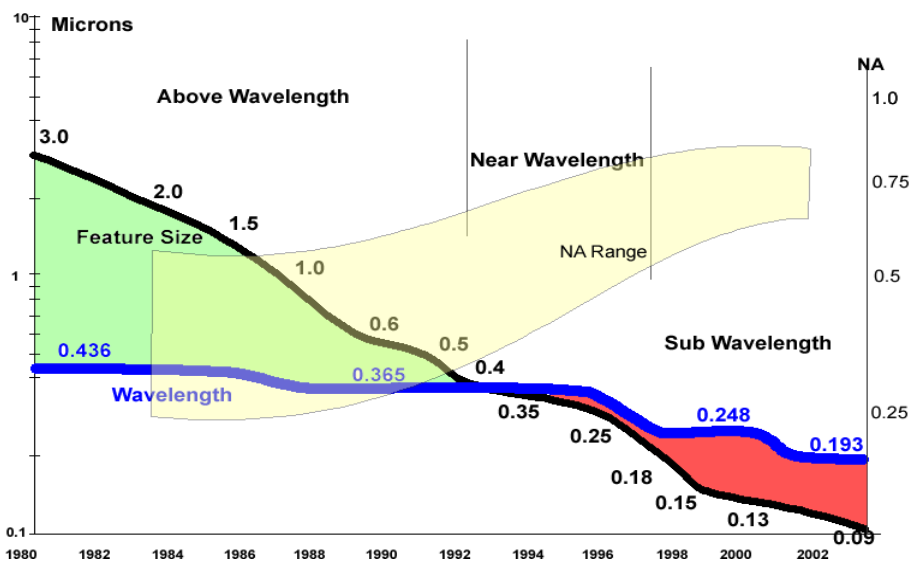
S. Brueck (University of New Mexico), argued that much more could be achieved with optical lithography. Enough, in fact, to reach the scaling limits of current materials and device designs. In particular, he pointed to laser interference patterns with very small pitches, immersion lithography (increasing the index of refraction *n* by 50%), and processing tricks. For example, after putting lines onto a resist, treatment with oxygen could thin them well below the limits achieved with lithography alone. In 2003, Intel's most advanced processes used this technique to achieve critical 90 nm spacings.

Fundamental Limits to Scaling

To a large extent, the operation of Moore's Law during the period 1965-2003 had been driven by MOS scaling, pushing the limits of optical lithography, and ever cleaner clean-rooms. Each of these processes did have some fundamental limits. The most severe were those of the transistor itself.

One such limit was the thickness of the oxide insulating barrier between the gate elec-

Figure 10
Evolution of Photolithography



trode and the silicon material beneath. By 2003 this layer was only a few atoms thick. Making it thinner ran into the essential granularity of matter—it was composed of atoms which were not divisible. Oxide layers of 1 or 2 atoms would also be susceptible to quantum tunneling—they would no longer insulate.

A second limit was that scaling was being achieved by increasing the concentration of dopants (e.g., boron, phosphorus). As these concentrations reached very high levels, there were no longer enough spaces left in the crystal for new dopants, so no new conductors (electrons or holes) were added.

A third factor was that the scaling rules depended upon reducing the operating voltage in proportion to the scale. But voltages less than about 0.25v appeared unachievable unless devices were super-cooled. At normal temperatures, such small voltages began to compete with background electron “noise” and ceased to define logical levels of “1” and “0.”

Despite these limits, many technologists had faith that “something would happen.” At a very general level, Moore’s Law could be taken to refer to computing power and cost. In such terms, many believed that it would continue to operate despite apparent limits imposed. If there were limits to scaling MOS devices, then other kinds of devices would be found. If there were limits to lithography, then other ways of manipulating matter would be developed.

Exhibit 1
Step-by-Step Fabrication of a CMOS Inverter

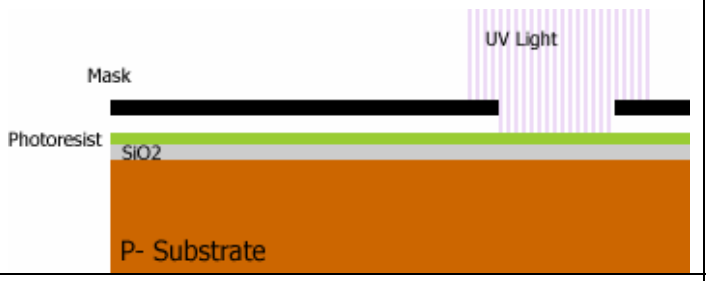
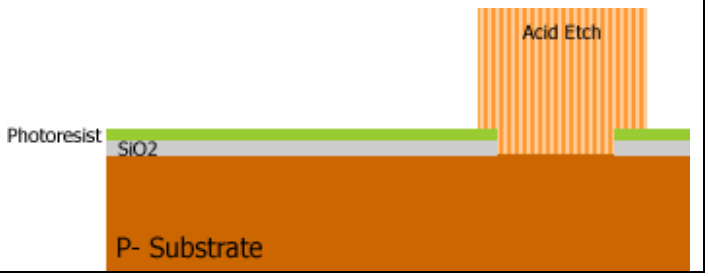
1. A wafer is received from the manufacturer. It has been produced with a light doping of p-type elements.	
2. The wafer is exposed to hot oxygen. The oxygen combines with the silicon to grow a layer of SiO2 (glass) in the surface. The layer is said to be "grown" because it expands as it forms.	
3. A layer of positive photoresist is created by spinning the wafer (3000rpm) with a drop of resist in the center.	
4. A mask is positioned and uv light shines through to soften the photoresist material wherever it strikes. The diagram focus in on one CMOS pair and omits the optical lens structure of the lithography system.	
5. The wafer is removed from photolithography and the photoresist is developed. The gap in the resist exposes the underlying SiO2 layer.	
6. Etch (hydroflouric acid) removes the SiO2 beneath. After the etch, the remaining photoresist is removed ("ashed").	

Exhibit 1 (continued)
Step-by-Step Fabrication of a CMOS Inverter

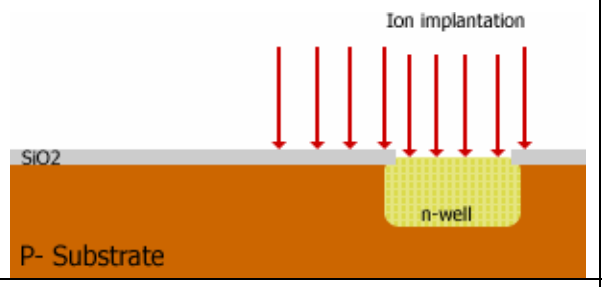
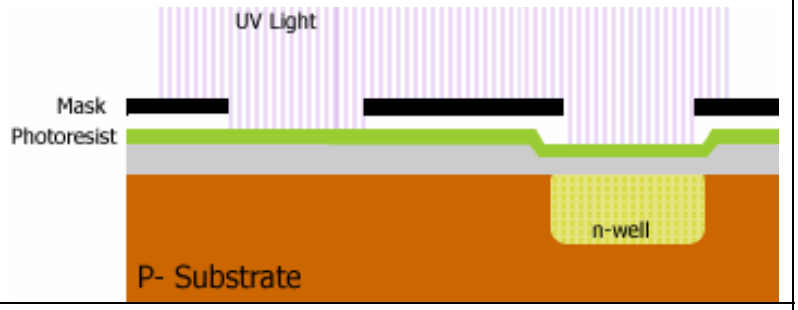
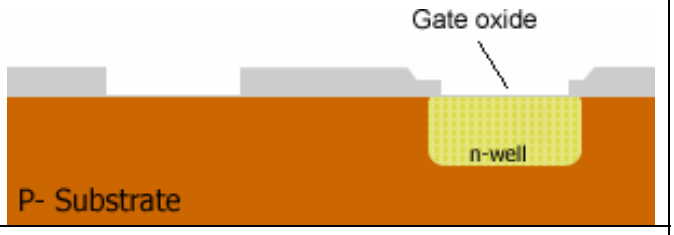
7. Phosphorus ion implantation: In this step the etched SiO2 acts as a mask protecting most of the silicon from the ion stream. The phosphorus ions penetrate the exposed silicon, creating a deep n-type region called a "well." This well later becomes the effective substrate for one of the two CMOS transistors.	
8. More hot oxygen thickens and grows SiO2 layer, recovering the n-well..	
9. Photoresist is spun on surface. A mask defining the active areas of transistors is positioned and the resist is exposed to uv light.	
10. SiO2 is etched away under exposed regions. Then, a very thin layer of SiO2 is regrown---the gate oxide layer	
11. A layer of polysilicon is deposited onto the whole wafer.	

Exhibit 1 (continued)
Step-by-Step Fabrication of a CMOS Inverter

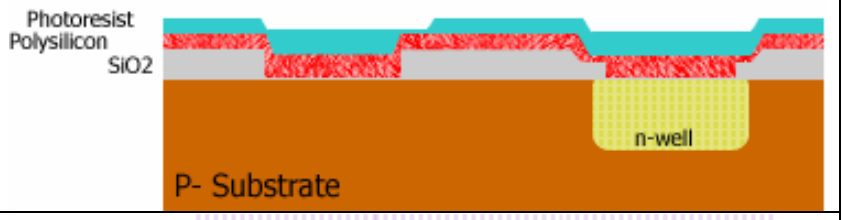
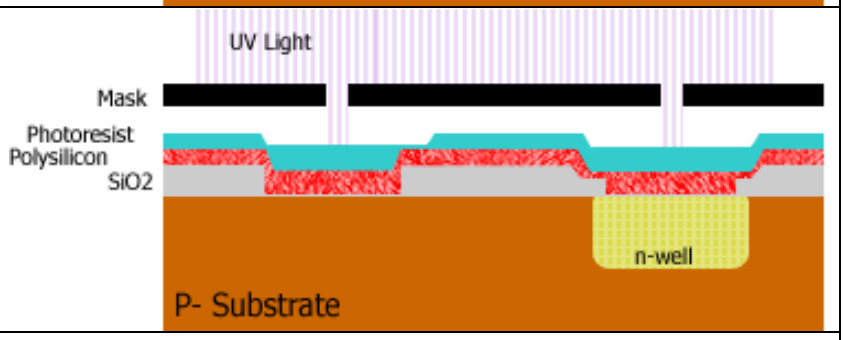
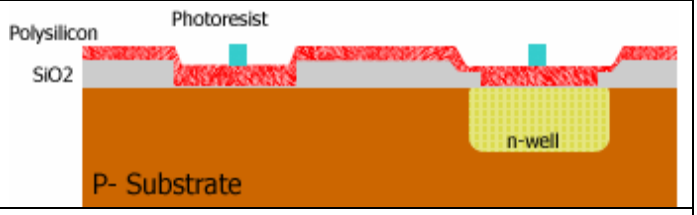
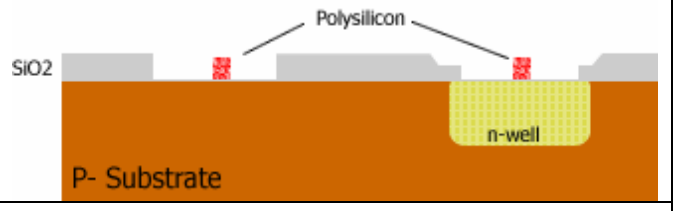
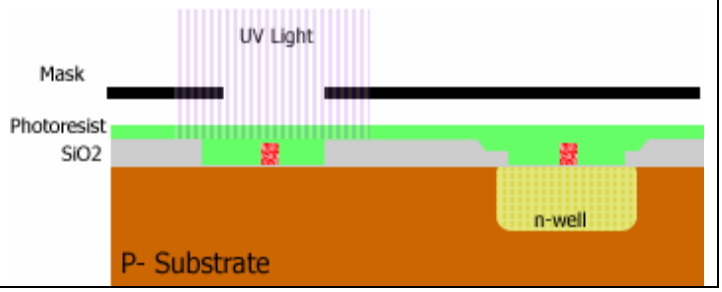
12. A negative photoresist is spun onto the surface of the whole wafer. The resist is "negative" in that it hardens, rather than softens, where exposed to light.	
13. A mask defining the gate regions is positioned and the resist exposed to the uv light that passes through.	
14. Photoresist is developed and removed except where exposed.	
15. The polysilicon is etched away except where protected by the resist. After removing the resist, the gates have been formed: a conducting layer of polysilicon over a thin layer of SiO2, over a doped substrate. A thin layer of SiO2 remains over the active areas.	
16. Photoresist is spun over wafer and a mask applied which defines the active area of the left-side transistor.	

Exhibit 1 (Continued)
Step-by-Step Fabrication of a CMOS Inverter

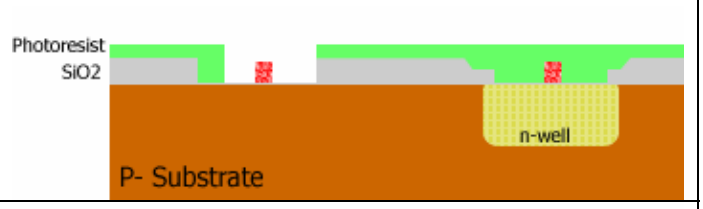
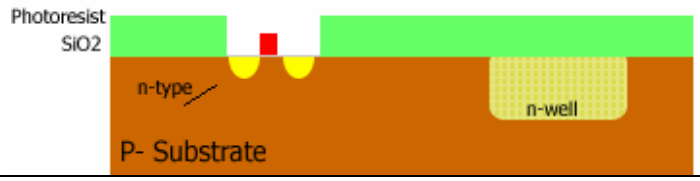
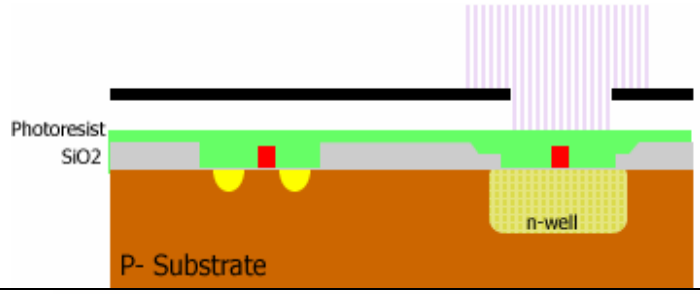
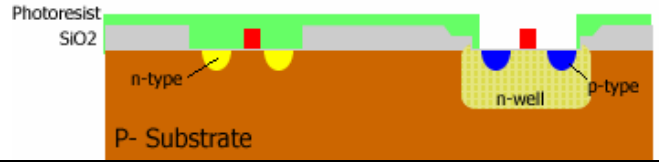
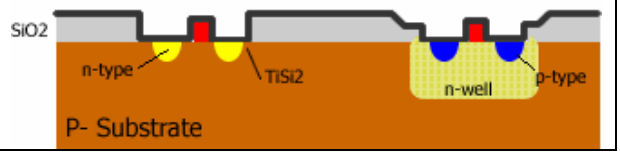
17. The photoresist is developed, exposing the active area on the left. A thin layer of SiO2 remains.	
18. High concentration n-doping (arsenic) penetrates the SiO2 layer. The resist and the polysiicon layers protect the rest of the wafer. This forms the basic structure of a transistor: two n-doped regions separated by a p-doped region.	
19. Photoresist is stripped and reapplied. The right-side active area is masked and exposed to define the region for p-type doping.	
20. The photoresist is developed and boron implantation creates p-type regions in the n-well, forming the structure of the complementary MOS transistor.	
21. The photoresist is removed. A thin layer of SiO2 is etched away, exposing the bare silicon above the doped regions. Titanium is deposited on the whole wafer by sputtering. Where the titanium touches silicon it forms TiSi2, a good conductor. Where it touches SiO2, it forms TiN, a poor conductor.	

Exhibit 1 (continued)
Step-by-Step Fabrication of a CMOS Inverter

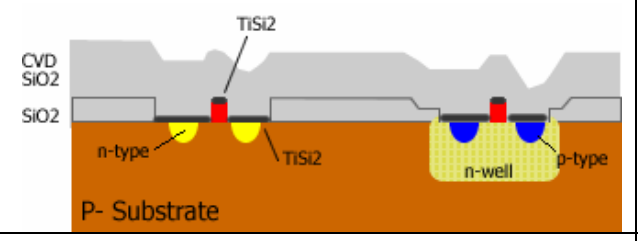
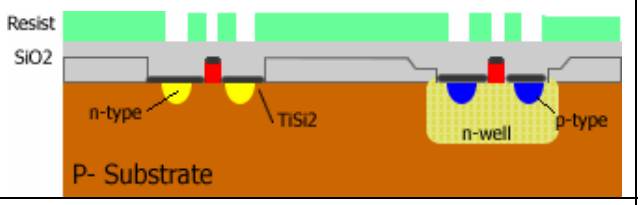
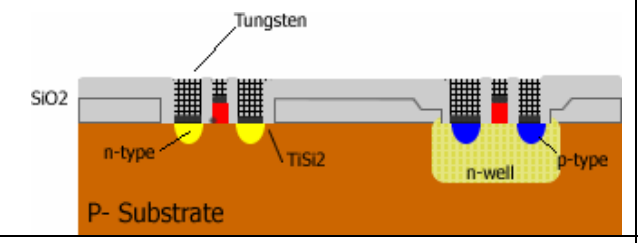
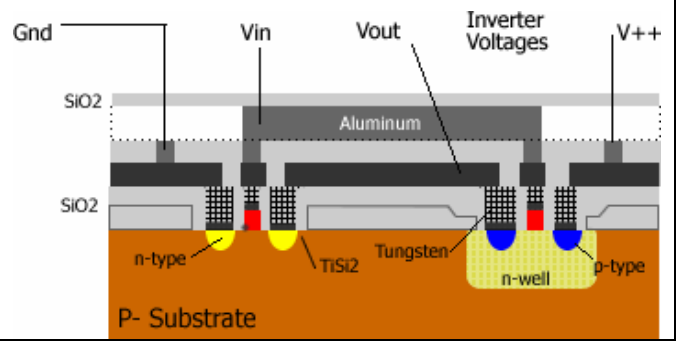
22. Photolithography and etching steps (not shown) remove the TiN, leaving conducting caps over the polysilicon gates and the p-type and n-type implants. After forming these contacts, the whole surface is covered with a layer of SiO2 insulator using chemical vapor desposition (CVD).	
23. The uneven SiO2 layer is smoothed, a process called planarization, by either mechanical polishing or resist etchback. Photoresist is spun onto the surface and patterned by photolithography. The wells in the resist define the positions of tungsten plug contacts.	
24. The SiO2 is etched down to the TiSi2 (not shown). Then, a relatively thick layer of tungsten (W) is sputtered over the entire wafer (not shown). The surface is again planarized, leaving the tungsten only in the etched holes in the SiO2.	
25. A layer of aluminum is deposited over the entire wafer. Photolithography defines cuts in the aluminum. SiO2 insulator is applied to the entire wafer, filling the cuts in the aluminum. The wafer is planarized. This defines the second metal layer of interconnects. More metal layers are added as necessary. A final finish of SiO2 is added to protect the wafer. The illustration shows to complementary MOS transistors connected to form a CMOS inverter.	

Exhibit 2
Integrated Circuit Cost Structures

		Year	1988	1988	2002	2002
		Product	DRAM 1 Mbit	Pentium II	DRAM 256 Mbit	Pentium 4 512KB Cache
A		Logic Transistors (millions)	0	7.5	0	41.9
B		Memory Transistors (millions)	1.024	0	256	13.1
C		Total Transistors (millions)	1.024	7.5	256	55
D		Wafer Diameter (mm)	150	150	200	200
E		Feature Size (nm)	1000	800	130	130
F		Wafer Starts/Week	5600	4500	6900	6900
G		Number of Masking Steps	10	17	25	23
		Cost of Processed Wafer ($/Wafer)				
H		Raw Wafer	40	60	60	85
J		Labor	132	163	102	174
K		Supplies	116	70	261	522
L		Equipment	132	497	962	1030
M		Facilities	12	38	93	89
N		*Total*	400	828	1478	1900
P		Die Size (sq-mm)	29	294	55.1	145.8
Q		Defect Density (per sq cm)	1.5	0.4	0.2	0.35
R		Cluster Parameter	3	1	2	1
S		Wafer Exclusion (mm)	2	2	3	3
T						
U	GrossDie()	Gross Die	551	48	510	185
V		Test/Control Sites	5	0	5	3
W	U-V	Gross Available Die	546	48	505	182
X	(1+P*Q/R)^-R	Probe Yield	67%	46%	90%	66%
Y	int(p*n)	Good Die	366	22	453	120
Z	N/Y	Cost/Good Die ($)	1.09	37.64	3.26	15.83
a		Package Cost ($/die)	0.15	2	0.6	8
b		Packaging Yield	0.98	0.99	0.99	0.99
c	int(a*b)	Good Packages	358	21	448	118
d		Final Test Cost ($/die)	0.16	0.16	0.15	0.5
e		Final Test Yield	0.9	0.95	0.99	0.95
f	int(c*e)	Good Chips	322	19	443	112
g	(N+Za+dc)/f	Cost/Good Chip ($)	1.59	42.31	4.09	25.87
h	g/B	Cost/Mbit ($)	1.59		0.016	
	g/C	Cost/M Transistors ($)	1.59	5.64	0.016	0.470

Notes to Exhibit 2

Costs have been estimated by combining technical information on devices with a number of industry sources, including Intel, Merrill Lynch, Goldman Sachs, IC Knowledge, SemaTech's "Wafercost Calculator," interviews with Mr. Howard Dicken, and the "Spreadsheet Model for Fab Economic Analysis," created by Leachman, Ding, and Sato-Misaw's UC Berkeley.

The yield equation used in Exhibit 2 was that given by (1) in the text.

The DieCount() function calculates the number of die that fit on a wafer. This function is function is implemented in VBA as follows:

```
Function DieCount(height As Double, width As Double, diameter As Double) As Integer
' (c) 2003 Richard P. Rumelt
'This function computes the number of die that can be fit onto a wafer. It takes the
'height and width of a single die and the diameter of the wafer as inputs and returns an
'integer count.
Dim h As Double
Dim w As Double
Dim row As Integer
Dim columns As Integer
Dim rowmax As Integer
DieCount = 0
h = height
w = width
If h > w Then
   h = width
   w = height
End If
If w < diameter Then
   rowmax = Int(Sqr(diameter ^ 2 - w ^ 2) / h)
   For row = 1 To rowmax
      columns = Int(Sqr(diameter ^ 2 - (row * h) ^ 2) / w)
      DieCount = DieCount + columns
   Next row
End If
End Function
```

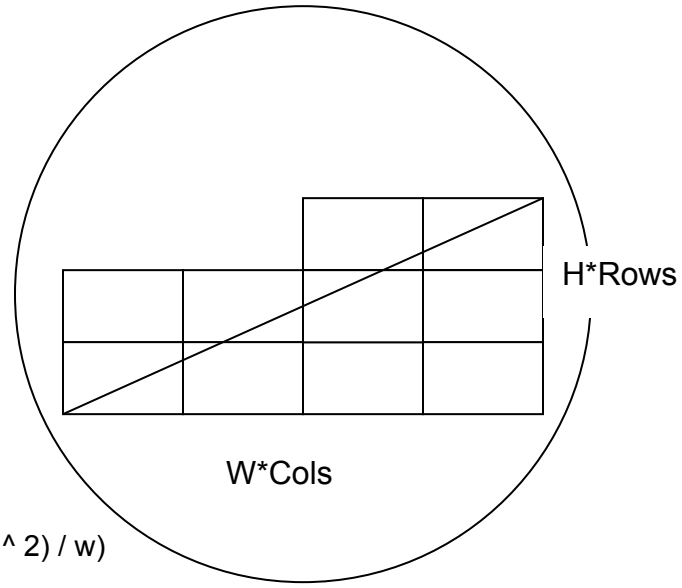


Exhibit 3
Die Area Computations

Year	2002	2002
Product	DRAM 256 Mbit	Pentium 4 512KB Cache
Die Area Computations		
DRAM Transistor Area		
Cell area factor	7	
Cell area (sq-microns)	0.1183	
Cells (million)	256	
Total DRAM Cell Area (sq-mm)	30.2848	
Logic Transistor Area		
Cell area factor		80
Cell area (sq-microns)		1.352
Cells (million)		41.944
Total Logic Cell Area (sq-mm)		56.71
SRAM Transistor Area		
Cell area factor		16.8
Cell area (sq-microns)		1.352
Cells (million)		13.056
Total SRAM Cell Area (sq-mm)		17.65
Total Active Cell Area (Sq-mm)		74.36
Silicon Efficiency	0.55	0.51
Die Area (sq-mm)	55.1	145.8

Notes to Exhibit 3

Cell area factors were multipliers that related the square of feature size to the area taken up by a transistor. Thus, if the feature size was F microns, and the cell area factor was 20, then the area taken by one transistor was $20F^2$ square microns. There were 1 million square microns in 1 square millimeter, so 1 million transistors would take up $20F^2$ square millimeters.

The active cell area on an integrated circuit were surrounded by a relatively empty margin which housed the metal leads running out to the terminals on the edge of the chip. Often, one-half of the area of the chip was devoted to this margin. The proportion of the chip used for actual active devices was termed the *silicon efficiency*.

DRAM memory normally used 1 transistor per bit. The actual data was stored as a charge on a capacitor. SRAM memory was much faster, but required 3 transistors per bit of storage. The fast cache memory included on microprocessors and logic chips was generally SRAM. A 512KB L2 cache required 0.512*8*3 = 12.3 million transistors. The 13.056 transistor count for SRAM on the P4 (above) included smaller caches as well, but excluded the instruction pipeline storage on the chip.

Exhibit 4

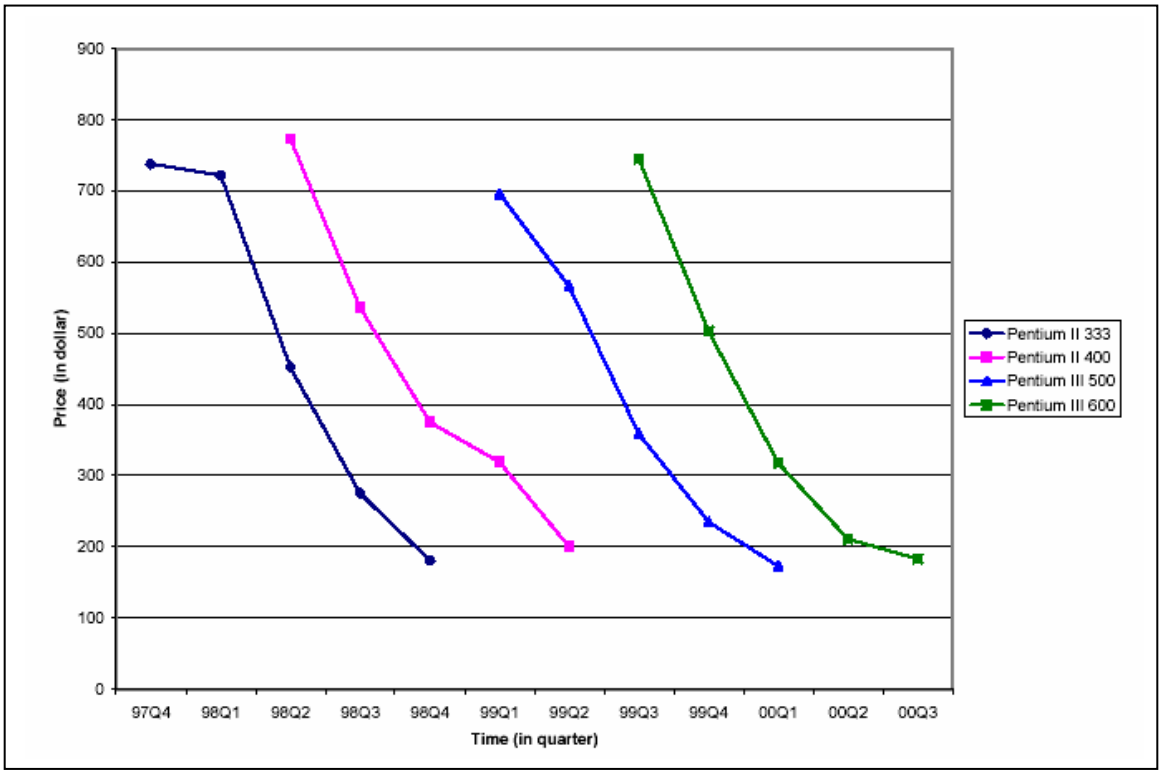
US ISO 14644-1, *Cleanrooms and Associated Controlled Environments*

CLASS	Number of Particles per Cubic Meter by Micrometer Size					
	0.1 um	0.2 um	0.3 um	0.5 um	1 um	5 um
ISO 1	10	2				
ISO 2	100	24	10	4		
ISO 3	1,000	237	102	35	8	
ISO 4	10,000	2,370	1,020	352	83	
ISO 5	100,000	23,700	10,200	3,520	832	29
ISO 6	1,000,000	237,000	102,000	35,200	8,320	293
ISO 7				352,000	83,200	2,930
ISO 8				3,520,000	832,000	29,300
ISO 9				35,200,000	8,320,000	293,000

The older cleanroom nomenclature dealt in 0.5 μm particles per cubic-foot. There are 35.2 ft3 per cubic-meter. Under the old system, ISO Class 3 was "Class 1" in that it had 35.2 particles per cubic meter, or 1 particle per cubic foot. Similarly, ISO Class 4 was called "Class 10," ISO Class 5 was called "Class 100," and so on.

Exhibit 5

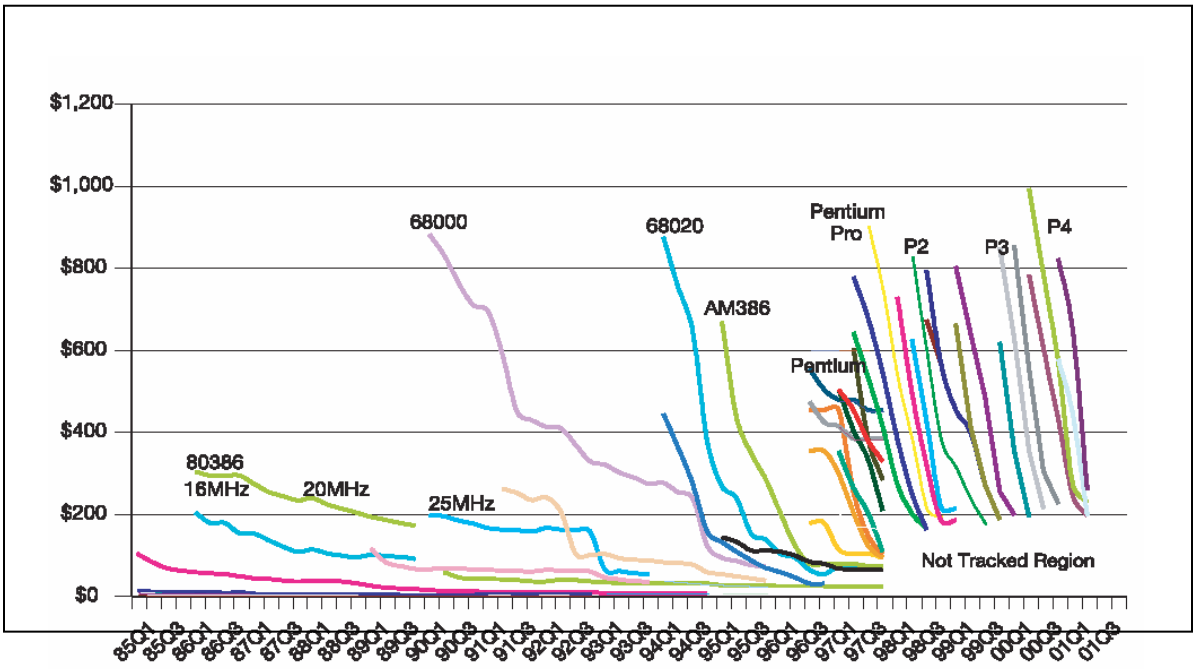
Microprocessor Model Price Trends



Source: Minjae Song - Measuring Consumer Welfare in the CPU Market. – 2002 and Microdesign Resources.

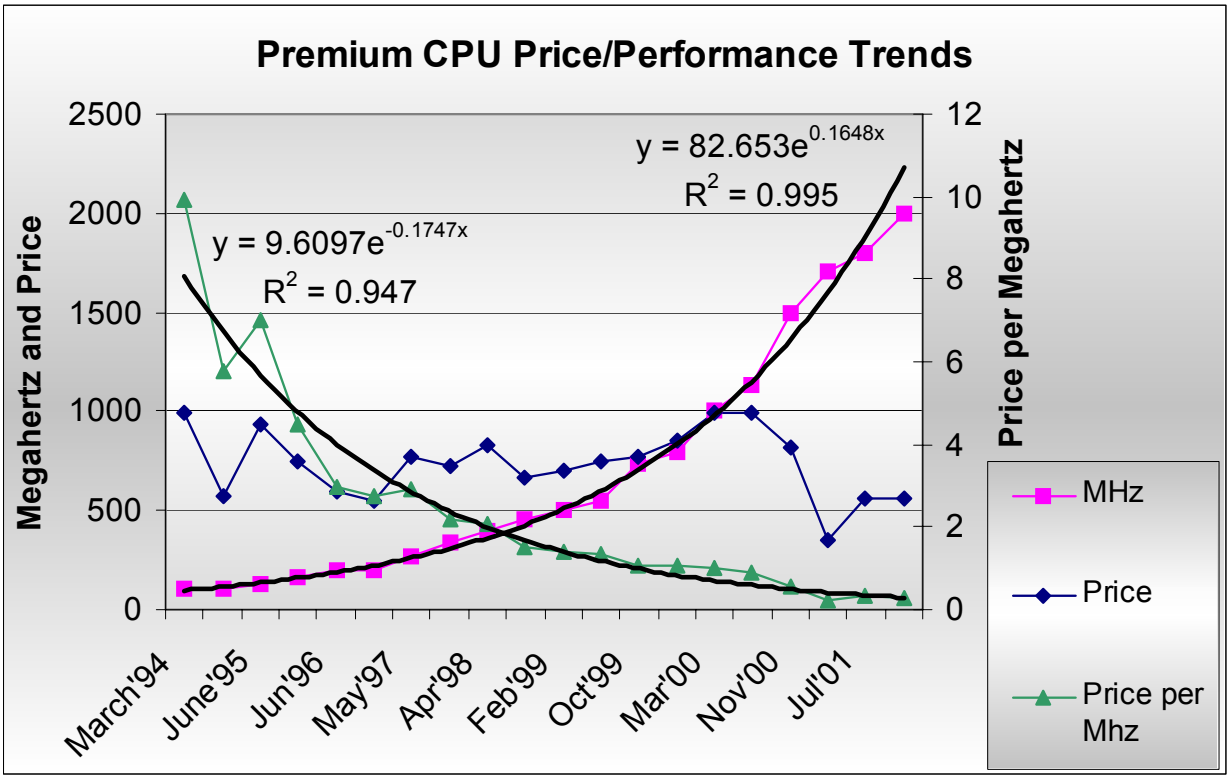
Exhibit 6

Microprocessor Lifecycles



Source: Dataquest and Microdesign Resources

Exhibit 7



Source: Intel

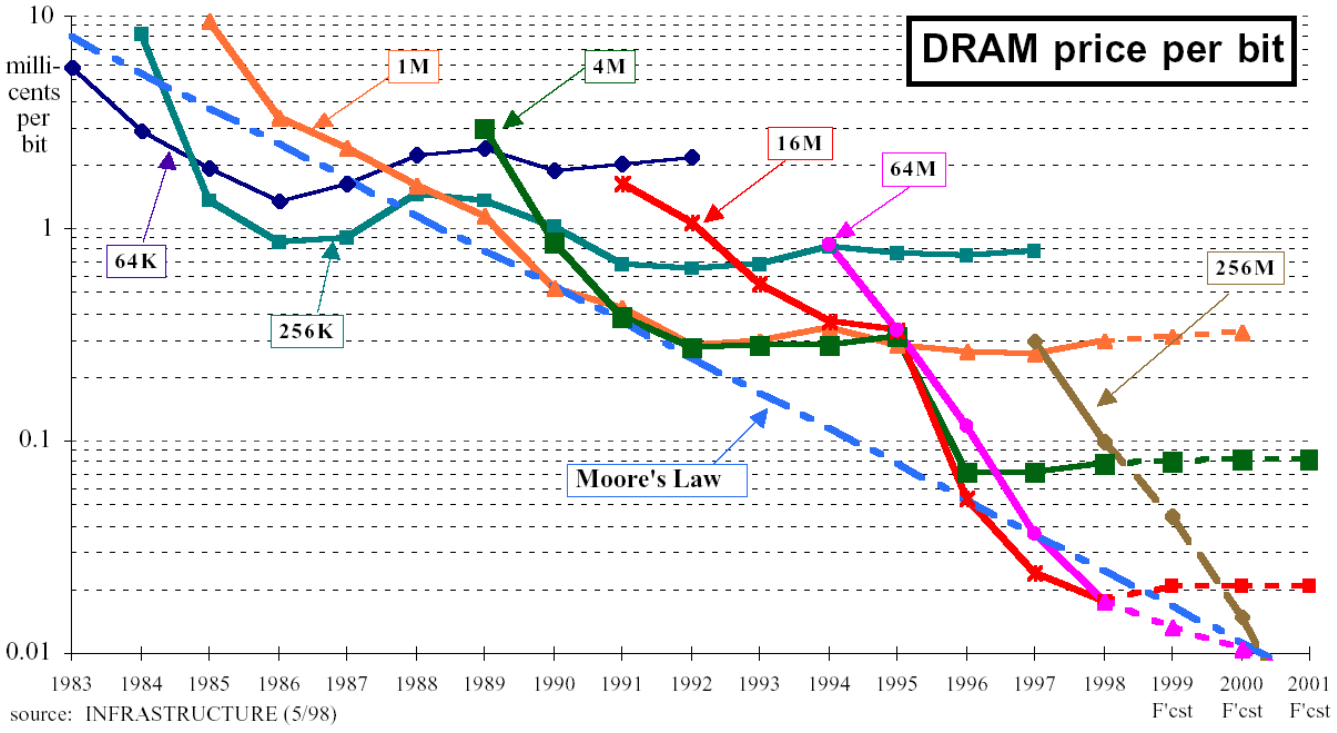



Exhibit 8
DRAM Price Trends